

Pixel identification in an image using Grover's algorithm

A Project Dissertation

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Abstract

Quantum Computing offers an entirely new way of doing computation governed by the rules of quantum mechanics like Superposition and Entanglement. These rules allow us to do computation over all the possible states simultaneously. Hence, offering exponentially higher computation power than the present classical computers. Quantum computing algorithms are entirely different from classical algorithms due to quantum parallel computing derived from quantum state superposition and entanglement, has natural advantages over classical image processing. Grover algorithm is a quantum based search algorithm used to find the correct answer from an unsorted database by computing on all the inputs simultaneously. Thus, giving us a quadratic speed-up of order $O(n)^{1/2}$ in comparison to the classical algorithm which offers speedup with order $O(n)$.

We used Grover algorithm for identifying the black pixel in a (2x2) classical image by first converting it into a quantum state and then running the Grover algorithm for identifying the pixel with 0 value maximum gray-scale intensity. This technique has applications in areas like steganography offering data encryption between users, image segmentation etc.

Key-words: Quantum Computation , Qubits, Quantum image, Grover algorithm

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1. History of Quantum computing

Quantum mechanics is the study of behaviour of microscopic systems. The idea of using its laws for doing the computation in an entirely different and amazing way has been hiding under from the field's early days. The underlying properties of quantum mechanical systems like the ability to be in superposition and entangled state forms the bedrock of this new and powerful way of doing computation[1]. The real trick goes in building a system that could be easily manipulated and measured with a desired accuracy. Richard Feynman a theoretical physicist is often honored with the beginning of the idea of quantum computers, several other researchers anticipated this idea. In 1979 a paper submitted by a young physicist, at Argonne National Labs named Paul Benioff, entitled "The computer as an actual system" he portrays an infinitesimal quantum mechanical hamiltonian model of computers as addressed by Turing machines. Benioff showed a theoretical based quantum computing model and suggested that it is possible to build such a computer. The computation process of such kind of a computer is described by a definite state of a system whose time evolution or processing is guided by the hamiltonian of the system. Thus, in principle a Turing machine could be thought of as set of pure states having definite phase relations with one another and whole system being guided by the Hamiltonian i.e. as the calculation progresses goes on these states are evolving according to the hamiltonian of the system. The theoretical models about such kind of computing machines at least brings up the idea that such kind of coherent machines can possibly be built and used for computational purposes[2]. Yuri Manin in his book Computable and Non-Computable in 1980 also presented the core idea of quantum computing. In 1981, theoretical physicist Richard Feynman presented a famous lecture "Simulating Physics with Computers". In this lecture, he explained the inadequacy of classical

systems. He mentioned that a quantum mechanical system can't be simulated by using a classical system and if we want to do it then we better do it quantum mechanically. His famous quote during the talk he gave is[1]:

...nature isn't classical, dammit, and if you want to make
a simulation of nature, you'd better make it quantum
mechanical, and by golly, it's a wonderful problem,
because it doesn't look so easy...

Based on the characteristics of quantum systems he then mentioned some of important characteristics that a quantum computer should possess so that it is useful in doing complex computational tasks[3]. At that time, however, the idea of building these kinds of computers and then using these to do any computational task was not clear to both Feynman and the physics. Once Manin, Benioff, and Feynman opened the doors, a new field of research emerged and many people began to ponder over the ideas of building such computers and algorithms that could be possibly be run on them. David Deutsch, a physicist at Oxford, came up with a more profound and practical framework for quantum computing in his 1985 paper. Deutsch talks in detail about the nature of quantum algorithms and how well can they be run on a quantum[14]. He anticipated that using the technology one day quantum computers will become reality and be able to solve real time problems.

2.Quantum Physics and Computation

At the point when we consider information in our every day life we consider information to be a theoretical numerical idea. What information is, and how it is evaluated, is an extremely expansive idea. For the time being, we return to our instinctive comprehension of the idea of data. Whatever information is ,for it to be

valuable (which is the thing that is required) it should be stored in some actual medium which can be controlled utilizing some physical processes that submit to laws of physics. In other words whatever the information is, it is physical in nature. This suggests that the capacities of any information processing machine are administered by the laws of physics[1] . Hence, physics plays a significant part in directing the working of such systems and it gets sensible to study information processing and specifically computation in the light of these laws of physics. Till the end of the 20th century, the laws of physics were believed to be what we presently call classical physics. Newton's laws of motion and Maxwell's theory of electromagnetism appeared to be finished in clarifying and anticipating the behaviour of such systems with exceptional exactness and precision. As researchers began analyzing marvels on nuclear scales, they showed some weird behaviors. It was found that a few investigations disagreed with the forecasts of classical physics. The consequences of these examinations appeared to be nonintuitive when seen from the laws of classical physics. These examinations included perceptions of wonders that arise at subatomic scale were not available in the times of Newton or Maxwell. The exceptional work of Planck, Bohr, de Broglie, Schrodinger, Heisenberg, in discovering these strange laws lead to the advancement of another hypothesis of physics which came to be known as 'Quantum physics' that appeared to clarify the behavior of such systems. Newton's and Maxwell's laws were discovered to be a guess to this more broad hypothesis of quantum mechanics. The classical estimation of quantum mechanics appears to hold up very well in clarifying the behaviour of plainly visible articles like planes, footballs, planets, or even particles. However, on the quantum size of subatomic particles i.e 'quantum size' of individual atoms, electrons, and photons, the traditional guess appears to be wrong , and the hypothesis of quantum mechanics should be considered to clarify these unusual practices[2]. A probabilistic Turing

machine is a machine verifiably reliant upon classical physics. We could fabricate such a machine with the assistance of generally huge actual segments, and every one of the parts of playing out a calculation and the conduct associated to it very well may be anticipated precisely by the laws of classical physics. One of the significant classes of tasks which are utilized to comprehend the actual frameworks is to mimic and study the evolution of these actual physical systems with the end goal that their behavior is perceived. Be that as it may, when we attempt to utilize our modern day classical computers to simulate these systems whose behaviour is unequivocally quantum mechanical appears to be an impossible task. Numerous physicists including Richard Feynman saw that we don't appear to do so proficiently utilizing classical systems since, we need to manage numerous factors of the framework and to register every one of the factors we appear to require high computational power[1]. Any attempt to reproduce the results of a nonexclusive quantum mechanical system on a probabilistic Turing machine appears to be basically unthinkable. Feynman proposed that it is conceivable to design a computer that could exploit the laws of quantum mechanics[2], that is, a computer whose evolution is unequivocally directed by quantum mechanical laws. Considering the above perception, apparently we were unable to utilize a probabilistic Turing machine for all intents and purposes to reenact such a computer. David Deutsch, who gave the principal formal model of a quantum Turing machine just as the quantum circuit model[16]. Deutsch showed that a potential model of computational machine could be made whose working depends expressly on the standards of quantum mechanics which is fascinating in itself. Is really stunning that numerous significant issues have been discovered that are simply conceivable to be tackled effectively just on a quantum computer, however no proficient answer for these issues is realized dependent on a probabilistic Turing machine[28]. This infers that this new hypothesis of quantum computation

has the huge potential and is offering high functional significance for the future just as of profound hypothetical interest for analysts[14]. In spite of the way that quantum calculation and quantum data offers numerous computational benefits there appear to be numerous difficulties for physicists, yet it's anything but a little ingenious what quantum calculation and quantum data offer to physics in the long run. We accept that similarly as we have figured out how to ponder computation, we can likewise figure out how to contemplate physics. While physics has been customarily viewed as a discipline focussed in on understanding the behavior of rudimentary objects and basic frameworks, many intriguing parts of nature emerge just when we go from simple systems to exceptionally convoluted ones[15]. Chemistry and engineering deal with such intricacy somewhat, yet regularly without consideration of wider application. Among the contributions of quantum computing and information is that new devices are accessible for analyzing the whirlpool between the simple systems and the generally complex systems. The utilizations of these ideas is as of now yielding new speculations into the world of physics. We trust that this viewpoint will help us in a thorough comprehension of the physical world far superior to previously and bloom sooner rather than later to come.

3.CLASSICAL INFORMATION

Quantum computing is a totally better approach for processing information. Thus, conventional strategies for computing that we are utilized to and acquainted with are alluded to as traditional data[14]. Classical Information Theory is the numerical hypothesis of data preparing errands like storage and transmission of data. The most essential piece in which this data is evaluated is known as a bit, which can be

used to addresses a yes-no response to any question[19]. To address this numerically, we utilize the way that we have to deal with a system having two-level and decide to address the information utilizing base 2 or binary numbers. Similarly as a binary number can be either 0 or 1. Likewise, bit can expect either among these two values. Genuinely we can carry out a bit with the assistance of an electrical circuit whose one end is at the ground (binary 0) and the other say +5 volts (binary1). As an initial step in getting to know the binary world, we should figure out how to check utilizing base 2. Before doing that, we need to realize the number of bits needed to address something. This can be resolved in the following manner: Suppose that a quantity can expect one of 'm' various states. Then, at that point, the number of bits needed to encode this data is given as:

$$2^n \geq m$$

for some n. The smallest 'n' for holding the above equation discloses to us the number of bits that are needed to address or encode these 'm' various states . To perceive how this functions, assume that we need to address the numbers 0, 1, 2, 3 in binary. We have four things, and

$$2^2=4.$$

Consequently at any rate two bits are needed to address these four numbers. To address the numbers 0 through 7, we require atleast three bits as $2^3=8$.

4. Quantum information

Quantum computation is the combination of three different branches of science and these are classical information, quantum physics and computer science[16].Theory and methods of various branches of maths and science are found useful in quantum computation, thus, are applied in this newly emerging and eye catching and

absorbing field[28]. The branches whose applications are in quantum computation are Abstract algebra, Linear algebra, Quantum statistical physics, Tensor theory, etc. Quantum computation proves how the different aims of information processing can be realised to get efficient speed for storing and processing data or information. The underlying laws of quantum theory lead us to realise the benefits of quantum computational theory as compared to the theory of classical information. In quantum computation, while processing data, information is stored as a physical system's quantum state[16]. Quantum theory is based on certain postulates. However, these postulates hold for only closed or isolated quantum systems (systems which are not interacting to the outer environment). In the real physical universe, totally isolated systems are not found at all. Real world systems are always interacting with the environment. Thus they are open, not closed. In quantum information theory, this interaction is called noise. This noise is always perilous for the information stored or is processing and gives us inaccuracy or error. This makes the system decoherent and it can lead to the total destruction of information in the system[28]. Quantum systems are very sensitive to these errors and make it necessary to understand the causes of this noising process and control these interactions while designing these quantum systems to make them highly efficient. Hence it's important to study the nature and behavior of open quantum systems.

Quantum bits

A bit or a classical bit in classical computational theory is the most basic unit of information analogous to its quantum counterpart known as the quantum bit which is called qubit in brief[19]. Let me try to explain the features of qubits by making a comparison to its classical counterpart. The benefits, by treating a qubit an abstract

entity, one can get in hand is that we are free to build a quantum computational theory that is independent of any practical system for its practical realisation. To the question how a real qubit is defined the answer would be that it is as, similar to a bit (classical) having two states 0 and 1 representing, for example, high and low voltages, a qubit can also represent two energy levels as two states. The two states are represented as $|0\rangle$ and $|1\rangle$ which are corresponding to the classical states 0 and 1. These notations $|\rangle$ are designed by P. Dirac and are called Dirac notations. The usefulness of a quantum bit over a classical bit is that a quantum bit can also represent superpositional states as a linear combination of the $|0\rangle$ and $|1\rangle$ states[28]. Which is the implementation of "Superposition Principle" the basic postulate of quantum theory[28]. The state of the system at that time can be written as:

$$|\psi\rangle = a |0\rangle + b |1\rangle$$

The coefficients 'a' and 'b' in the above equation represent complex numbers, in general. Although one can also take them as real numbers and doesn't lose much of generality. To understand a qubit mathematically, a qubit can also be considered as an element or a vector in a 2D Hilbert space[2]. The states (classical) $|0\rangle$ and $|1\rangle$ represent what are usually called computational basis states. The computational basis states make the orthonormal basis of the system. We can look at a classical bit to decide if it is in the state 0 or 1 like computers do constantly while recovering the substance of their memory[19]. Maybe amazingly, we can't straightforwardly look at a qubit for deciding its quantum state, and look for the values of 'a' and 'b'. All things considered, quantum mechanics discloses to us that we can just gain significantly more confined data about the quantum state. At the point when we do a measurement on a qubit we get just one of the results for example we get either the outcome $|0\rangle$ or $|1\rangle$ with likelihood of getting 0 given by

$|\alpha|^2$, and the likelihood of getting $|1\rangle$ given by $|\beta|^2$. Normally, $|\alpha|^2 + |\beta|^2 = 1$ i.e. the sum of probabilities should be equal to one as per Born rule. Mathematically, one may decipher this as the state of qubit is normalized to length 1. In this way, when all is said, a qubit's state can be thought as a vector of unit length living in a two-dimensional vector space. This contrast in the undetectable condition of a qubit and the perceptions that can be made lie in the core of quantum computation and quantum data hypothesis. Considering the greater part in our theoretical models of the world, there is a connection among the components of the reflection and the present reality, similarly as a planner's arrangements for a structure are in correspondence with the last structure. Anyway this kind of direct correspondence is missing in quantum mechanics which makes it hard to examine the conduct of such frameworks; nonetheless, there exists an indirect correspondence, for qubit states can be controlled by specific methods and changed in manners that lead to measurement results that rely unmistakably upon the various properties of the state. Consequently, these quantum states have genuine, experimenally undeniable outcomes, which we will see are important to the force of quantum computation and quantum data.

The capacity of a qubit to explore the superposition principle and be in mixed states is nonintuitive according to our 'common sense' notion of systems that we encounter in the world[2]. A classical bit can be believed to be as practically equivalent to a coin[16] which can either yield two outputs as tails or heads up. In defective coins, there might exist moderate states like having it balancend on an edge, which can be ignored in the best case. On the other hand, a qubit can exist in a limitless number of potential states that can be shaped by the direct superposition of $|0\rangle$ and $|1\rangle$ until it is estimated[28]. Since, when a qubit is measured, it just at any point gives '0' or '1' as the measurement result – probabilistically. For instance, a qubit can be in the state

$$\frac{1}{\sqrt{2}} [|0\rangle + |1\rangle]$$

which, when measured, gives the result 0 fifty percent ($|1/\sqrt{2}|^2$) of the time, and the result 1 fifty percent ($|1/\sqrt{2}|^2$) of the time[2]. Despite showing this sort of peculiar behaviour, qubits are distinctly genuine, their reality and conduct have been broadly approved by tests, and a wide range of actual frameworks can be utilized to acknowledge qubits in useful applications. To get a substantial feel for how a qubit can be acknowledged by and by it could be useful to show a list of the manners in which qubits could be acknowledged[28]. One can thought of two different polarization directions of a photon with one polarization direction $|\downarrow\rangle$ serving as state $|0\rangle$ and other polarization direction $|\uparrow\rangle$ as state $|1\rangle$ or the alignment of a nuclear spin in a uniform magnetic field or as being the two states of an electron orbiting a single atom as shown in fig.1 below

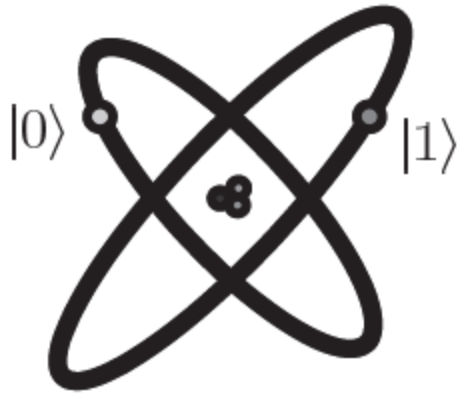


Fig.1 Realization of a qubit in two electronic levels in an atom.

Representation of state vector

The state of a system in quantum mechanics is something containing the total information at a particular time which is a vector contained in Hilbert space known as a state vector. Since this vector is of length 1 unit we can represent these states

on a circle of unit sphere otherwise called the Bloch sphere. Consider the condition of a qubit as:

$$|\psi\rangle = \alpha |0\rangle + \beta |1\rangle \dots\dots\dots (1)$$

Because $|\alpha|^2 + |\beta|^2 = 1$ (Born rule), from equation (1) we have :

$$|\psi\rangle = e^{i\gamma} \left(\left[\cos \frac{\theta}{2} |0\rangle + e^{i\phi} \sin \frac{\theta}{2} |1\rangle \right] \right)$$

where θ , ϕ , and γ are real numbers, the factor of $e^{i\gamma}$ (global phase) can be completely ignored because when we consider the observables it has no effect on them, so using it in equation(1), we can write,

$$|\psi\rangle = \left[\cos \frac{\theta}{2} |0\rangle + e^{i\phi} \sin \frac{\theta}{2} |1\rangle \right]$$

The unique values of θ and ϕ characterize an extraordinary point on the unit sphere in three dimensions, as displayed in Fig(2). This sphere is frequently called the Bloch sphere and it's anything but a helpful method by which the condition of single qubits can be pictured[27].

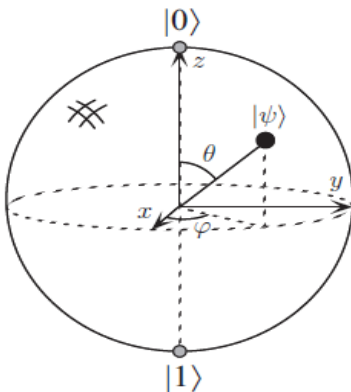


Fig 2(a) Bloch sphere representation of state of qubit.

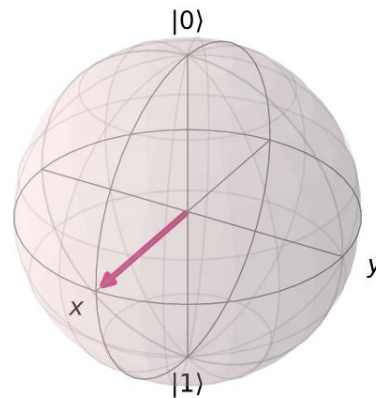


Fig 2(b) Representation of even superposition of $|0\rangle$ and $|1\rangle$ state on Bloch sphere

It frequently fills in as a magnificent bedrock for understanding quantum computation and quantum data. A considerable lot of the activities that are

continued single qubits can be perfectly imagined on the outside of Bloch circle. In any case, it should be remembered that this instinct is restricted to single qubits simply because there is no basic speculation of the Bloch sphere known for multiple qubits.

Quantum measurement

Quite possibly the most troublesome and questionable issues in the entire of quantum mechanics is the thing that we call measurement issue. The measurement issue in quantum mechanics emerges because of the association of classical system with a quantum system and it is one of the incredible inexplicable riddles in the entire of quantum mechanics[15]. The issue emerges on the grounds that quantum mechanics gives the aftereffects of an investigation in probabilistic way for various potential results in a trial. It gives no component by which the genuine, at last noticed outcome, occurs. Obviously, probabilistic results include in numerous places of classical physics also, however all things considered, likelihood enters the image just on the grounds that we have a lacking quantity of information to make any unmistakable expectation in regards to the systems along these lines, we need to think about the probability theory. On a fundamental level, that missing data is there to be discovered, it is only that getting to it could be a reasonable difficulty. Interestingly, there is no such 'missing data' for a quantum system, what we see is all that we can get, even on a fundamental level, however there are speculations that say that this missing information dwells in hidden variables[11]. The von Neumann hypothesis plainly expresses that as an estimation is completed the immediate result of it is that the condition of the system goes through an irregular change, for example $|\psi\rangle \rightarrow |q_n\rangle$ if the outcome q_n is acquired on carrying out a measurement of an observable Q . This immediate change in state from superposition to definite one is known as 'the collapse of the state vector'[28]. This

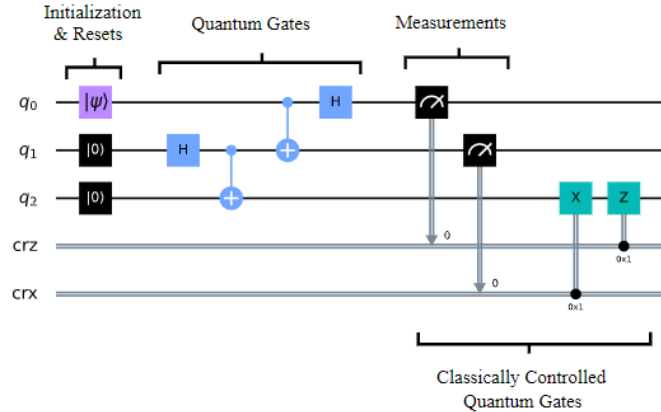
raises the feeling that the interaction of measurement fundamentally includes actual interaction with the system, that at long last, brings about a significant actual disturbance in the condition of a system – one second the system is in a state $|\psi\rangle$ (i.e. superposition of all possible states), the next it is forced into a state $|q_n\rangle$ (i.e. in one of a particular state)[16]. When a measurement process is carried onto a quantum system, we get a probabilistic outcome and the state of the system collapses from superposition of all distinct states to a definite state. So a measurement can reveal only one outcome at a time so in order for the answer to come correct, we design our algorithms in such a way that all the wrong answers are canceled out and only right answers remain, which can be done by using interference effects[31].

5. Quantum circuits

What is a Quantum Circuit?

A quantum circuit can be considered as a computational course comprising of lucid quantum procedure on quantum information, for example, qubits, and simultaneous real time classical computation[30]. It's anything but an arranged grouping including definite course of action of quantum doors, measurements and resets, which may all be molded on and use information from the ongoing traditional calculation.

A bunch of quantum gates is supposed to be universal if unitary transformation of the quantum information is productively approximated efficiently by a grouping of gates in the set. Any quantum program can be addressed by an arrangement of quantum circuits and non-simultaneous classical computation.



The circuit shown in fig (3) uses three qubits and two classical bits along with some quantum gates. There are four main components in this quantum circuit.

Initialization and reset

To begin with, we have to begin our quantum system from one of definite quantum states. This can be accomplished utilizing the introduction of the system in a definite state and reset activities. The resets can be maintained by a blend of single-qubit gates and concurrent real time classical computation that makes it sure whether we have succeeded in making the ideal state by measurements. The statement that q_0 into an ideal state $|\psi\rangle$ would then be able to follow by the operation of single-qubit gates.

Quantum gates

Second, we operate a grouping of quantum operations or tasks which control the condition of quantum bits as per computation. For this situation, we have applied single-qubit Hadamard (H) and two-qubit Controlled-X ($\oplus$) doors.

Measurements

This is the significant part as the information required can be extricated solely after this part .

A classical system deciphers the measurement of each qubit as traditional results (0 and 1) and stores them in the two classical bits.

6. Pauli spin matrices

The Pauli spin matrices are a bunch of three 2×2 complex matrices which are Hermitian and unitary. These matrices structure the premise of any 2×2 vector space and assume a significant part in understanding quantum computation. These are as:

$$\sigma_1 = \sigma_x = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$$

$$\sigma_2 = \sigma_y = \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix}$$

$$\sigma_3 = \sigma_z = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}$$

These matrices are named after the physicist Wolfgang Pauli. All these matrices are Hermitian, and along with matrix I (also considered as the zeroth Pauli matrix σ_0), the Pauli matrices structure a reason for the genuine vector space of 2×2 Hermitian matrices. This implies that any 2×2 Hermitian matrix can be described in as straight mix of Pauli matrices, with reasonable coefficients. Hermitian operators address the observables in quantum mechanics. Thus, Pauli matrices span the space of observables described in two dimensional complex space. Since our qubits lie in a 2D complex Hilbert space Pauli matrices assume a significant part in Quantum computation.

7. Quantum gates

Changes happening to the state of a system can be portrayed utilizing the theory of quantum computation. Comparable to the manner of building a classical computer using an electrical circuit comprising of wires and logic gates, in the same way quantum computer is build using a quantum circuit made up of wires and elementary quantum gates[14]. The function of such a circuit is to manipulate and control the quantum information. In the following part we depict few straightforward quantum operations or gates, and describe a few model circuits and their applications.

The Pauli Gates

Pauli-X gate

The X-gate can be described by the Pauli-X matrix:

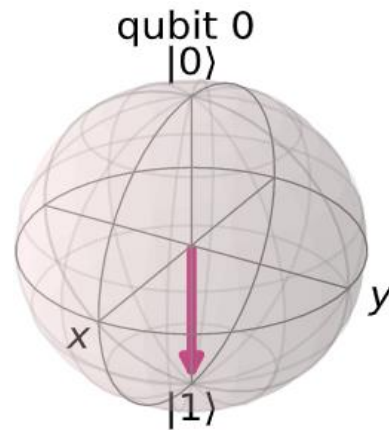
$$\sigma_1 = X = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix} = |0\rangle\langle 1| + |1\rangle\langle 0|$$

To see the effect of this gate on a qubit, we simply multiply the qubit's state vector $|0\rangle$ by the Pauli-X gate.

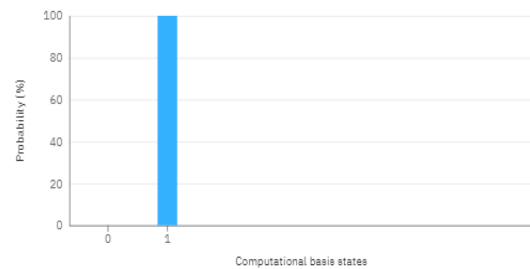
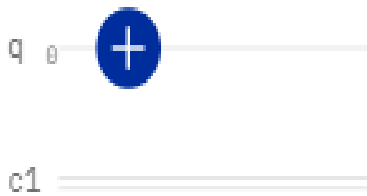
$$\sigma_1 |0\rangle = X |0\rangle = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix} \begin{bmatrix} 1 \\ 0 \end{bmatrix} = \begin{bmatrix} 0 \\ 1 \end{bmatrix} = |1\rangle$$



Fig.4(a) Circuit representation of X gate



(b) Rotation on Bloch sphere $|0\rangle$ to $|1\rangle$



Fig(4) Implementation of X gate on state $|0\rangle$ on real time quantum circuit simulator on IBM.

The X-gate flips between the states $|0\rangle$ and $|1\rangle$. It is clearly evident from the fig.4 that the state of the qubit is $|1\rangle$ as expected Fig4(b). This can be thought of as a rotation of Bloch vector by π radians around the x-axis of the Bloch sphere. Based on this application X-gate is also referred as a NOT operation because, it does something similar to the NOT gate in classical computation.

The Y & Z-gates

Additionally to the X-gate, the Y and Z Pauli matrices likewise go about as the Y and Z-gates on our qubits and, actually like X-gate they likewise perform turns by π around the y and z-axis separately of the Bloch sphere when applied on qubits.

$$\sigma_2 = Y = \begin{bmatrix} 0 & i \\ i & 0 \end{bmatrix} = -i |0\rangle\langle 1| + i |1\rangle\langle 0|$$

$$Y|0\rangle = i |1\rangle \quad ; \quad Y|1\rangle = -i |0\rangle$$

$$\sigma_3 = Z = \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix} = |0\rangle\langle 0| - |1\rangle\langle 1|$$

$$Z |0\rangle = |0\rangle \quad ; \quad Z|1\rangle = -|1\rangle$$

The X, Y & Z-Bases

Notice that the Z-gate when applied to states $|0\rangle$ and $|1\rangle$ appears to have no effect when applied to these states. This implies that these two states $|0\rangle$ and $|1\rangle$ form the eigen states of Z-gate. This is the reason we often refer the basis made by these two states as Z-basis. However there are also basis for X and Y gate, one important basis of X gate formed by the superposition of eigen states of the Z-gate which serves as the computational basis as it yields the true power of superposition of states $|0\rangle$ and $|1\rangle$. This basis also has two eigen vectors that span it. We call these two vectors as $|+\rangle$ and $|-\rangle$ and they are expressed as:

$$|+\rangle = \frac{1}{\sqrt{2}} (|0\rangle + |1\rangle) = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 \\ 1 \end{bmatrix}$$

$$|-\rangle = \frac{1}{\sqrt{2}} (|0\rangle - |1\rangle) = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 \\ -1 \end{bmatrix}$$

The basis formed by eigen states of Y-gate are less common in computation and are shown as $|i\rangle, |j\rangle$.

The Hadamard gate

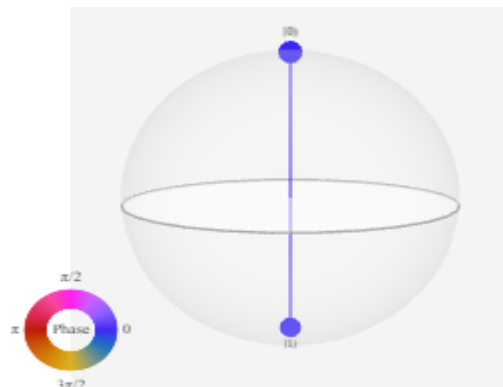
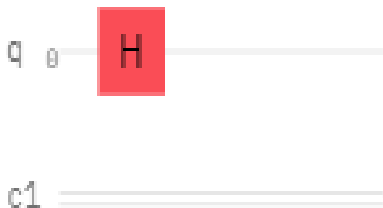
The H-gate or Hadamard gate is an essential quantum gate[30]. This gate permits us to transform our state vectors on the Bloch sphere by taking it from the poles and making a superposition of $|0\rangle$ and $|1\rangle$. It has the matrix

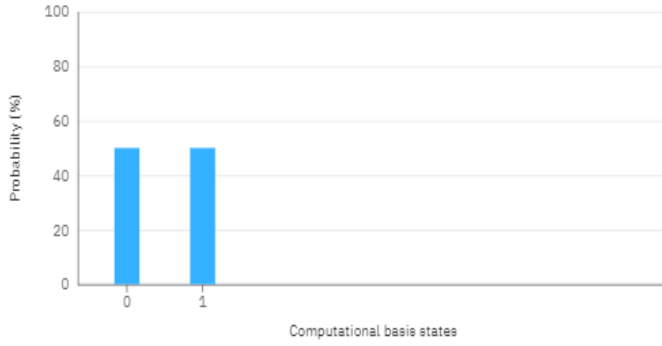
$$H = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix}$$

$$H|0\rangle = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix} \begin{bmatrix} 1 \\ 0 \end{bmatrix} = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 \\ 1 \end{bmatrix} = \frac{1}{\sqrt{2}} (|0\rangle + |1\rangle) = |+\rangle$$

$$H|1\rangle = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix} \begin{bmatrix} 0 \\ 1 \end{bmatrix} = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 \\ -1 \end{bmatrix} = \frac{1}{\sqrt{2}} (|0\rangle - |1\rangle) = |-\rangle$$

It is clearly seen that the application of Hadamard gate on qubits takes them from a definite state of either $|0\rangle$ or $|1\rangle$ to an infinite set of possible states given by the superposition of $|0\rangle$ and $|1\rangle$. The Hadamard gate can be used to switch from Z basis (i.e $|0\rangle$, $|1\rangle$) to the X basis (i.e $|+\rangle$, $|-\rangle$). So, Hadamard gate is very necessary for computation because it enables us to explore the superposition states of a system at the same time which is the reason of speedup of quantum computers. The circuit for Hadamard is shown as:





Fig(5) Implementation of H gate on state $|0\rangle$ and results based on IBM qiskit.

The Hadamard gate can be applied to more than one qubit. To apply it to more than one qubit we actually take the tensor product of it. The tensor product of n Hadamard gates is described as $H^{\otimes n}$ where, $n = 1, 2, 3, \dots$ is the number of qubits on which it is applied. In general, the application of $H^{\otimes n}$ to a product state with n copies of $|0\rangle$ is given as:

$$H^{\otimes n}(|0\rangle^{\otimes n}) = \frac{1}{\sqrt{2}^n} \sum_{x \in \{0,1\}^n} |x\rangle$$

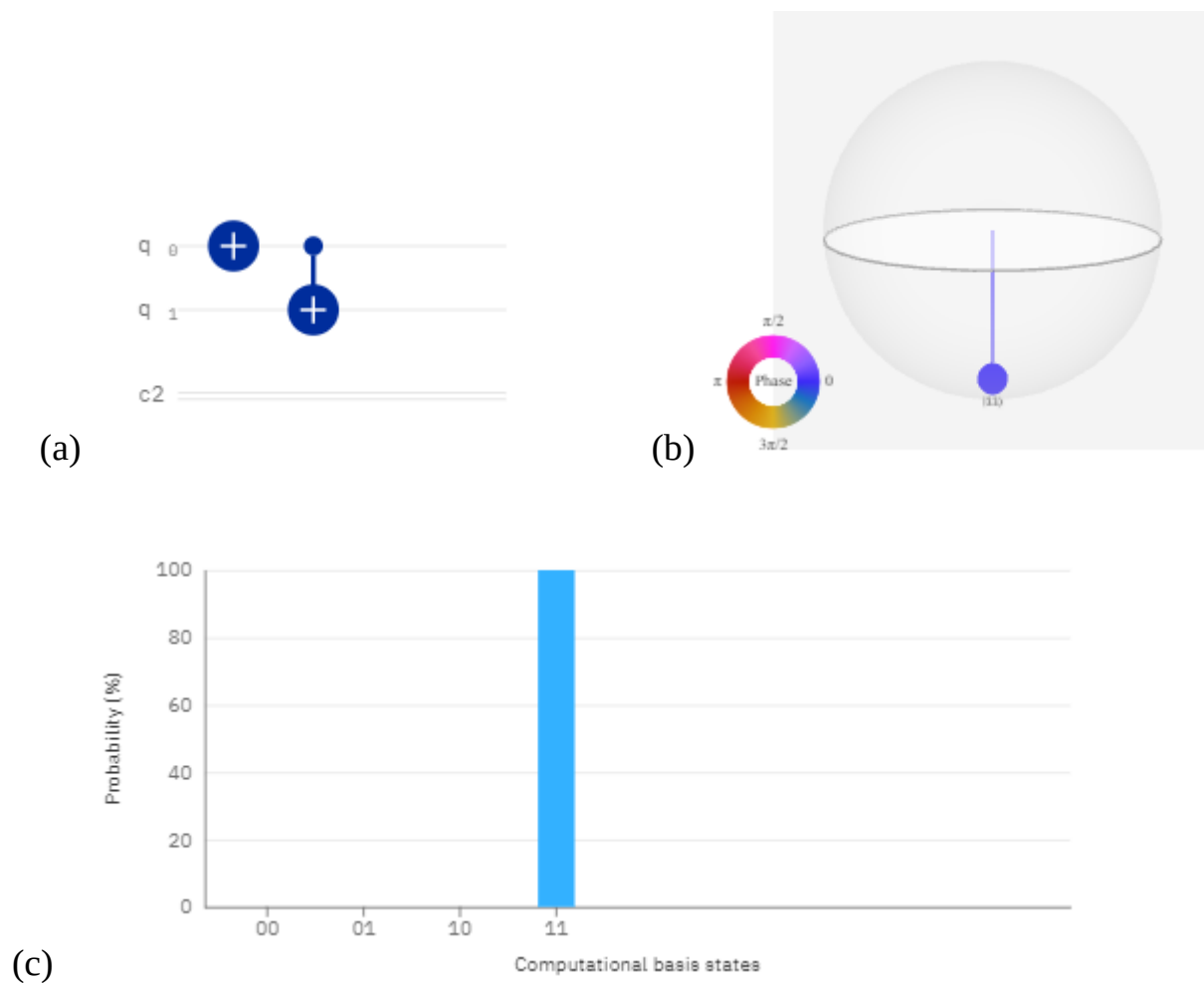
Where the factor at the start is just for the conservation of probability. (i.e the sum of all probabilities should be equal to 1).

The Controlled NOT gate (C -NOT & CC–NOT gate).

One more important quantum gate is the controlled not gate also known as C-NOT in short. This gate also has wide variety of applications in quantum computation like creating one of the bells states which are entangled states and are used for quantum teleportation[30].

The C-NOT gate is applied on two qubits so, it involves an operation onto two qubits, one of the qubits is called the control qubit and the other qubit is called as target qubit. Its function completely is characterized by the state of control qubit. It

applies C-NOT the NOT operation on target qubit depending on the state of the control qubit. If the control qubit is in state $|1\rangle$ the C-NOT gate applies NOT operation on target bit and if it is in state $|0\rangle$ it leaves the target qubit unchanged. The circuit diagram for C-NOT gate is shown in fig.6 below. The blob ($\bullet$) on the upper part is the control and the ($\oplus$) is the target qubit where X operation is carried out depending on target qubit.



Fig(6) Application of C-NOT gate in qiskit applied on state $|10\rangle$ with control being $|1\rangle$.

In Fig(6) above the circuit implementation of C-NOT gate is shown. (a) shows C-NOT gate applied for qubit 'q₀' and 'q₁'. A NOT operation is first carried on the qubit q₀ which is the control bit in our circuit, so its state changes from $|0\rangle$ to state $|1\rangle$. The output of the circuit is to be read in reverse order that is q₁ comes first and q₀ comes second at the output. The output states of the circuit are plotted in (c) where the state of qubit q₁ is flipped only when q₀ is $|1\rangle$.

8.Quantum algorithms

A quantum algorithm is a bit by bit procedure for tackling an assignment, similar to a classical algorithm. In Quantum computation every one of these means can anyway be performed on a quantum computer (i.e. we can always construct quantum equivalent of classical gates and circuits used in classical algorithms)[16]. The term quantum computation is typically utilized for those calculations which are innately quantum, or utilize some fundamental component of quantum mechanics like quantum superposition or entanglement[14]. Quantum algorithms are the arrangement of tasks that can be performed on a quantum computer and accomplish a speedup, or other productivity improvement, over any conceivable old style calculation[24]. The idea of using quantum mechanics for obtaining a solution to a problem with lesser time wholly depends on the efficiency of algorithm that we are using to obtain the result. Quantum information processing without efficient algorithm is a total chaos. In both the classical and quantum settings, the proficiency of a calculation is the deliberate of the quantity of rudimentary tasks utilized by that calculation in taking care of a particular issue[31].On the basis of quantum calculation, this can be approximated by using the quantum circuit model, in which a quantum circuit is considered to be a grouping of rudimentary quantum operations called quantum gates, being applied to small number of qubits. For comparing the time complexity of algorithms we

use $O(f(n))$ notation which gives us the idea about the number of evaluations taken by an algorithm in solving a problem. However, Problems which are not capable of being decided by using classical computers can't be decided either utilizing quantum computers[24]. What makes quantum algorithms fascinating is that they could possibly take care of certain problems quicker than classical computer in light of the fact that the quantum properties like superposition and entanglement that quantum computers use can't be used by classical computers. Researchers have come up with many efficient algorithms and the work for designing new possible algorithms is growing day by day. However there are some famous algorithms like Deutsch Jozsa algorithm [2](also known as DJ algorithm) which is used in finding whether a function is constant or balanced for the values of inputs. Shor's algorithm[8] given by Peter Shor for factoring the numbers which lies at the heart of our security systems. Shor's algorithm offers exponential speedup in finding factors than any best-known classical algorithm for factoring. Grover's search algorithm[9] by L.K.Grover for searching an entry in an unsorted or an unordered list. This algorithm has almost quadratic speedup than any classical algorithm for solving these kind of search problems.

Grover algorithm

Grover's algorithm demonstrates the practical capability of quantum algorithms over classical algorithms[10]. Grover algorithm is used to find answer to search problems by looking in an unsorted list. This algorithm can speed up the searching process in an unstructured dataset almost quadratically than any classical algorithm, however, its uses stretch out past that; it can fill in as an overall procedure to acquire quadratic speedup upgrades for various different algorithms by using what we call as amplitude amplification trick. This methodology could be utilized to speed up search algorithms by developing a "quantum oracle" that

recognizes some particular property from a set like distinguishing needle from hays. The needle and haystack are not required to be essential part of database. For instance, it very well may be utilized to look for two whole numbers $1 < a < b$ with the end goal that $ab = n$ for some number n , in this way playing out the work of a factoring algorithm[9]. Executing the quantum oracle can be decreased to developing a quantum circuit that flips an auxiliary qubit, q , if a function, $f(x)$, assesses to 1 for an input x . We utilize the term auxiliary or subordinate qubit to refer to some extra qubits that are utilized by the algorithm. The $f(x)$ is characterized by:

$$f(x) = 1 \text{ if } x = x^0$$

$$f(x) = 0 \text{ for all other } x$$

where $x = x_1x_2 \dots x_n$ are binary strings and x^0 is the particular string that will be found. It might appear to be dumbfounding from the start that an algorithm for finding x^0 is required if such a function can be developed. The key here is that $f(x)$ need just perceive x^0 – it is like the contrast between recording a condition and addressing a condition. For instance, it is not difficult to check if the result of ab is equivalent to n , however harder to factor n . Generally, Grover's algorithm can transform a discretionary capacity with double yields, if we have a quantum oracle that might actually carry out the capacity. Grover's calculation has been utilized, with suitable oracles, to tackle issues like discovering triangles in a diagram, discovering cycles etc[9]. For the examination of Grover's algorithm, the internals of the oracle is commonly viewed as a black box equipped for perceiving the answer for a problem. Often, the oracle operator for the problem at hand has to be constructed as a quantum circuit[18]. But, keep in mind that if an inefficient oracle is constructed this could nullify all practical advantages gained by using Grover's algorithm. The procedure of finding a winning state (say W) from a set of possible states can be obtained by Grover algorithm using the following steps:

State preparation

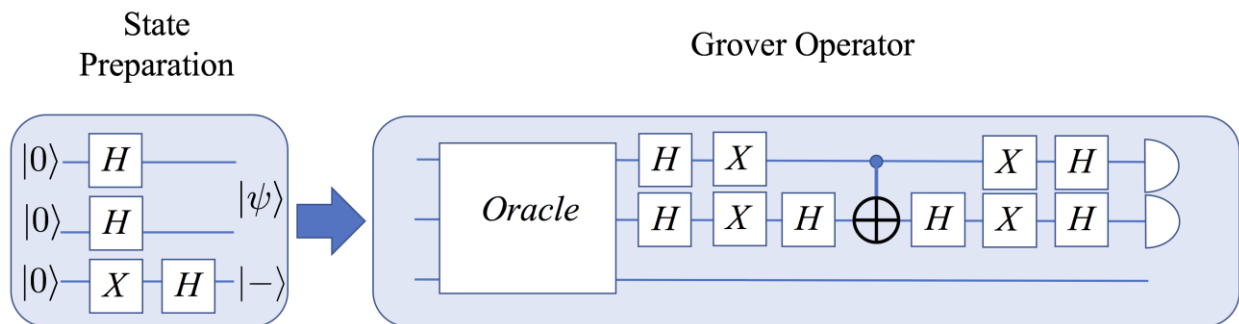
The first step of Grover algorithm starts with initializing the system from set of definite states of qubits. this is done by using the hadamard transform over all the qubits such that a uniform even superposition of states is formed containing all the possible answers (i.e. right answers as well as wrong answers)

Flipping of the winning state by Oracle.

An oracle is like a black box capable of recognizing any particular property of the system defined by some constraints on which it is applied. It does not calculates anything at all, but only says that whether any input given to it satisfies the constraints or not. If constraints are satisfied by some input it will recognize it and flip its state from $+$ to $-$ otherwise it will do nothing to it. Thus what oracle does actually is that it shifts the phase of winning state W with respect to other states of the system. In our case all the superposition states are given at the input of the oracle circuit and oracle recognizes the winning state ' W ' if it is present in the input by shifting its phase by ' π ' and if not then nothing is done to them. This depiction about the prophet without portraying how it functions practically speaking is fairly unique, and astounding in itself. It appears to be like the oracle definitely knows the response to the pursuit issue; what conceivable use could it be to have a quantum search calculation dependent on such oracle counsels? The appropriate response lies in the way that there is a differentiation between realizing the answer for any hunt issue, and remembering it; the critical point is that it is feasible to do the last without essentially having the option to do the previous.

Amplitude amplification by Diffuser

Since quantum mechanics is probabilistic and what we get at the output is a probabilistic answer so in order to get the right answer that is required at the output with high probability it is necessary to increase the probability of winning state to 1 and Make it 0 for all other states[10] . This is done by using the diffuser circuit which continuously flips the winning state about the average amplitude of all other states such that when the measurement of the system is done we get winning state with high probability. This can be done in a few number of steps depending on the number of input entries N and no of winning states. The advantage of this algorithm is that if there are more number of winning states the algorithm takes much less time then what is needed when only one winning state is searched.

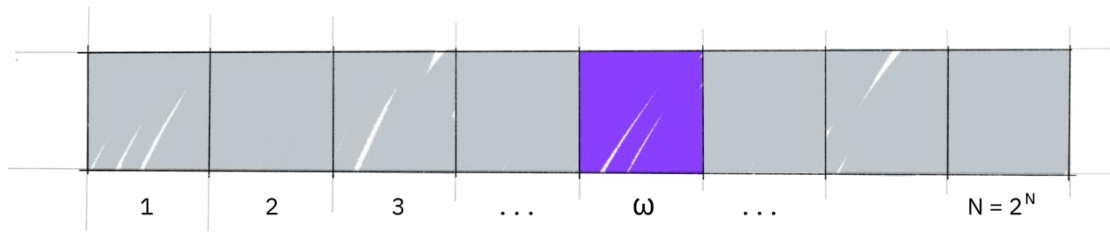


Fig(7) A schematic diagram of Grover's algorithm is shown. Note that in this case, one application of the Grover operator is performed. This is all that is necessary when there are only two bits in x , but the Grover operator should be applied repeatedly for larger problems

9.Geometrical interpretation of Grover algorithm

Our qubits lie in a 2D complex Hilbert space spanned by two linearly independent states it is possible to show them geometrically on a plane in which the two dimensions are described by state vectors and the operations can be shown as rotations on the plane.

Assume we are given an unsorted extensive collection of N items. Suppose among these N inputs we will look for a particular thing with some exceptional property ; we will call this one the entry as our winning state W . Consider everything in the list as a container of a specific color. Say all things in the list are dark aside from the vector W , which is purple.



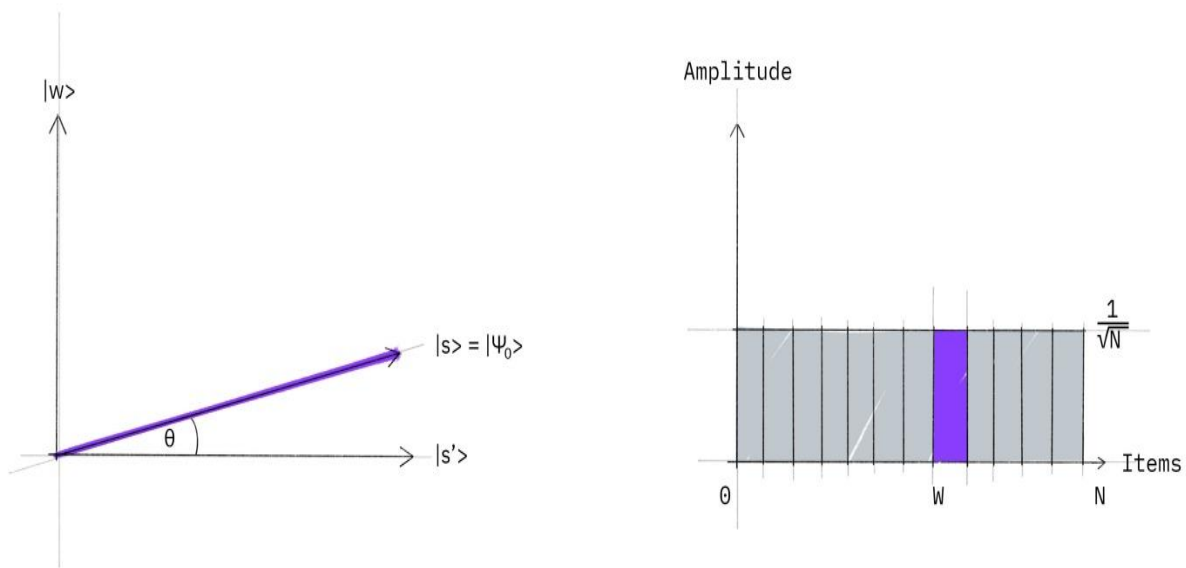
To obtain the winning state (i.e. the marked item) using classical computation, one would have to check all the inputs one by one in best case it will be present at the start and in worst case, at the last of the list in that case all $N=2^n$ inputs are to be checked. So, on average it takes about $N/2$ searches which is of the order $O(N)$. Using a quantum computer the solution to this problem can be found in roughly $\sqrt{N}$ steps. A quadratic speedup in fact fills in as an efficient device for discovering answers for search issues from long unsorted records. Furthermore, the working of algorithm doesn't rely upon list's inner construction, which makes it more broad general and it's anything but a quadratic quantum accelerate for some classical problems.

This algorithm has a pleasant mathematical understanding in terms two reflections, which create a rotation in a two-dimensional plane. The unique states we need to consult with are the winner state $|w\rangle$ and superposition state $|s\rangle$. The state $|s\rangle$ being our initial even superposition state. These two vectors are said to be spanning the whole of two-dimensional plane contained in the vector space C^n . The states $|s\rangle$ and $|w\rangle$ are not orthogonal because $|w\rangle$ state is also present in the superposition

state with amplitude $\frac{1}{\sqrt{N}}$ as well. So, we introduce an additional state $|s'\rangle$ being orthogonal to $|w\rangle$ and are spanning plane containing $|s\rangle$. The winning state $|w\rangle$ is obtained by removing $|s'\rangle$ from $|s\rangle$ and rescaling.

Step1: The amplitude amplification trick that grover algorithm utilizes starts in forming the superposition state $|s\rangle$, which can be easily formed by applying Hadamard transform to initial qubits as follows:

$$|s\rangle = H^{\otimes n} |0\rangle^n$$



The left side of the Fig shows a two dimensional plane being spanned by two orthonormal states $|w\rangle$ and $|s'\rangle$. The initial state of the system can be clearly described as a linear superposition of $|w\rangle$ and $|s'\rangle$ as:

$$|s\rangle = \sin\theta |w\rangle + \cos\theta |s'\rangle \quad \theta = \arcsin \langle s|w\rangle = \arcsin \frac{1}{\sqrt{N}}$$

$$|s\rangle = \frac{1}{\sqrt{N}} \sum_{x=0}^{N-1} |x\rangle. \quad \text{where } N = 2^n \in (0, 1)^n$$

In the left side of the Fig the bar graph of the amplitudes of the state is shown. It can be seen clearly that the probabilities of all the states is the same as $\left(\frac{1}{\sqrt{N}}\right)^2$.

Step2: The next step is to identify our winning state ' ω ' among all these 2^n states and flip its amplitude. This is done by applying the oracle function U_f applied to the state $|s\rangle$. The action of U_f is shown as:

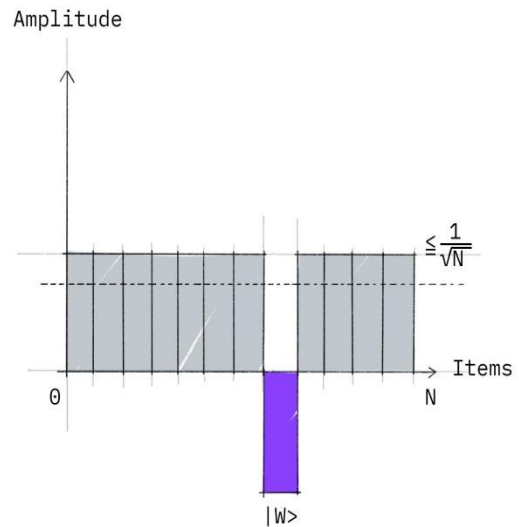
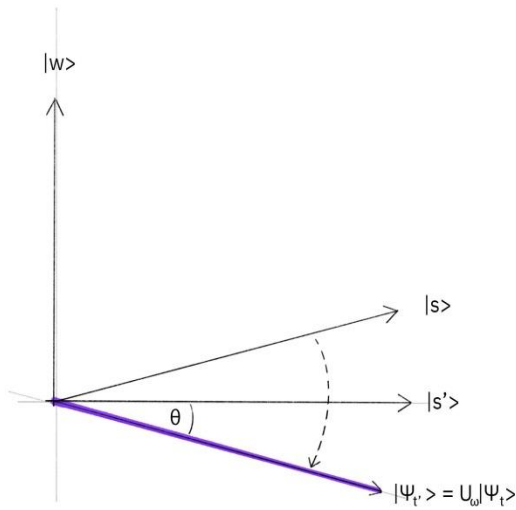
$$U_f |x\rangle = \begin{cases} -|x\rangle & \text{if } x = \omega \\ |x\rangle & \text{if } x \neq \omega \end{cases}$$

$$U_f |x\rangle = (-1)^{f(x)} |x\rangle$$

Where $f(x)$ is a function whose value is 0 for all $x \neq \omega$ (i.e. x is not a solution)and 1 only if $x = \omega$ (i.e. x is a solution).

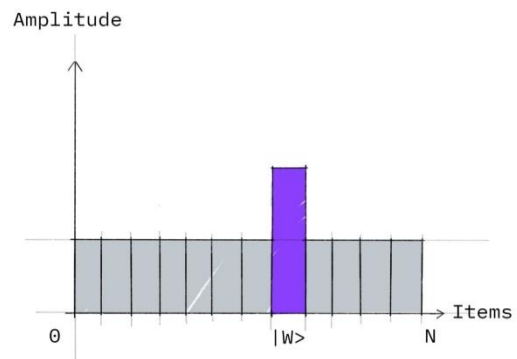
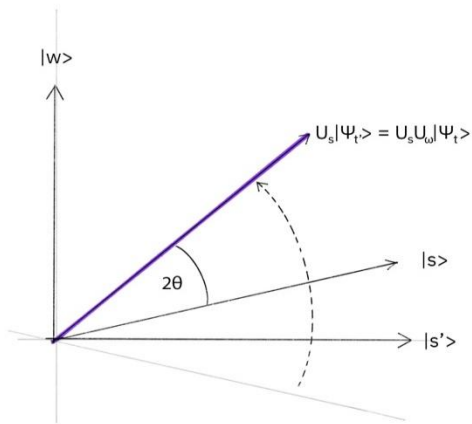
Geometrically the action of U_f can be shown as a reflection of the state $|s\rangle$ about $|s'\rangle$. This application of U_f on $|s\rangle$ results in the flipping of the amplitude of the $|\omega\rangle$ state, resulting in the decrease in the average amplitude of all states other states (indicated by a dashed line) has been lowered as shown in the fig below:

:



Step 3. The third step involves increasing of the amplitude of winning state and decreasing it for all other states. This is achieved by the Diffuser circuit which reflects the state $|w\rangle$ over $\sqrt{N}$ times. The step is carried out by applying U_s on $U_f|w\rangle$.

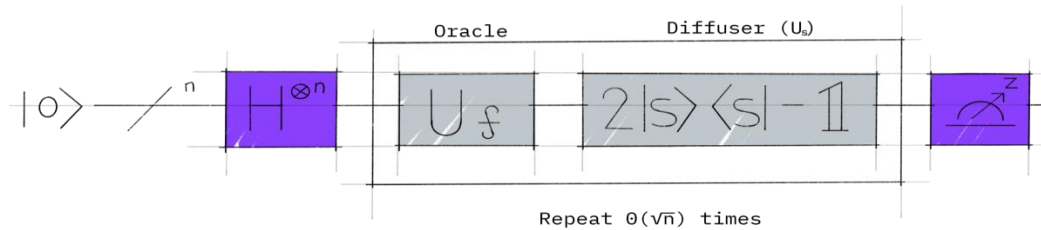
$$U_s = 2|s\rangle\langle s| - 1.$$



Two reflections consistently relate to a rotation[30]. The uses of $U_s U_f$ on the underlying state $|s\rangle$ turn the state $|s\rangle$ closer towards the vector $|w\rangle$ so that the probability at getting winner $|w\rangle$ when the measurement is completed is increased. The action of U_s can be perceived as a reflection being done about the normal average of any remaining 2^{n-1} states. Since the average amplitude in the principal reflection had been brought down, this changes the negative amplitude of $|w\rangle$ and boosts it to approximately multiple times its original, while it diminishes the amplitudes of any remaining states. We would then be able to again apply the stage 2 if necessary over and over with the end goal that there is greatest likelihood for state $|w\rangle$.

We now ask how many rotations or applications of operation U_s suffice to obtain the required output. This becomes more clear by taking a look at the amplitudes of the state $|\psi\rangle$. We can see that the amplitude of $|w\rangle$ develop directly as the number of applications $\sim \sqrt{N-1/2}$. Be that as it may, in the vector space's measurement a square root enters, since our main concern is with amplitudes not with the probabilities. Consequently amplitude, and not simply the likelihood, that is being enhanced during the technique.

In the case that there are M multiple solutions to a given problem, our algorithm takes roughly $\sqrt{N/M}$ rotations to obtain a solution, which is less than what is required when only one solution exist. This implies that the time complexity of the problem decreases as number of possible solutions contained within a data set increases which is another advantage of Grover algorithm. The application of Grover algorithm can be summarized in the following picture.



10. Two qubit Grover algorithm

Let's look at the grover algorithm when it is applied to two qubits (i.e $N = 2^n = 4$).

In this case only we have to apply only one rotation to get the required answer.

In the case $N = 4$.

$$\text{We have } \theta = \arcsin \frac{1}{2} = \frac{\pi}{6}$$

After t steps, we have

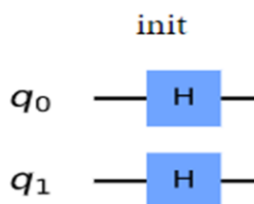
$$(U_s U_f)^t |s\rangle = \sin \theta_t |\omega\rangle + \cos \theta_t |s\rangle,$$

Where,

$$\theta_t = (2t+1)\theta.$$

In order to obtain $|\omega\rangle$ we need $\theta t = \frac{\pi}{2}$ which with $\theta = \frac{\pi}{6}$ inserted above results gives the value of $t = 1$. This implies that the diffuser circuit is to be applied only once to get the answer.

When we initialize the circuit for two qubits we get all the four possible states of a system that are $|00\rangle, |01\rangle, |10\rangle$, and $|11\rangle$. The initial state of the system $|s\rangle$ is given by the superposition of all the four states which is obtained by taking hadamard transform $H^{\otimes 2}$ over $|0\rangle^{\otimes 2}$:



$$|s\rangle = \frac{1}{2} (|00\rangle + |01\rangle + |10\rangle + |11\rangle).$$

Let's try our oracle for a particular state $|11\rangle$ (i.e. $|w\rangle=|11\rangle$).

The action of oracle U_f on $|s\rangle$ can be shown as follows:

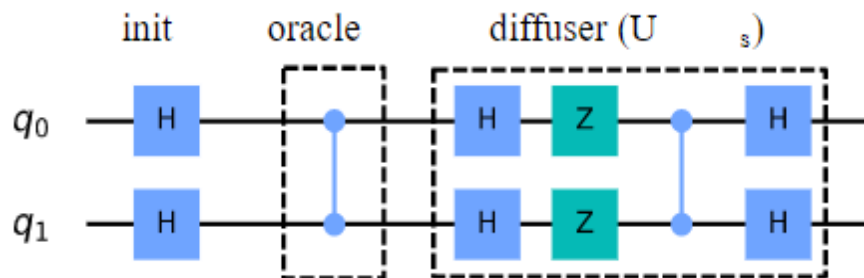
$$\begin{aligned} U_f |s\rangle &= U_f \frac{1}{2} (|00\rangle + |01\rangle + |10\rangle + |11\rangle). \\ &= \frac{1}{2} (|00\rangle + |01\rangle + |10\rangle - |11\rangle). \end{aligned}$$

or:

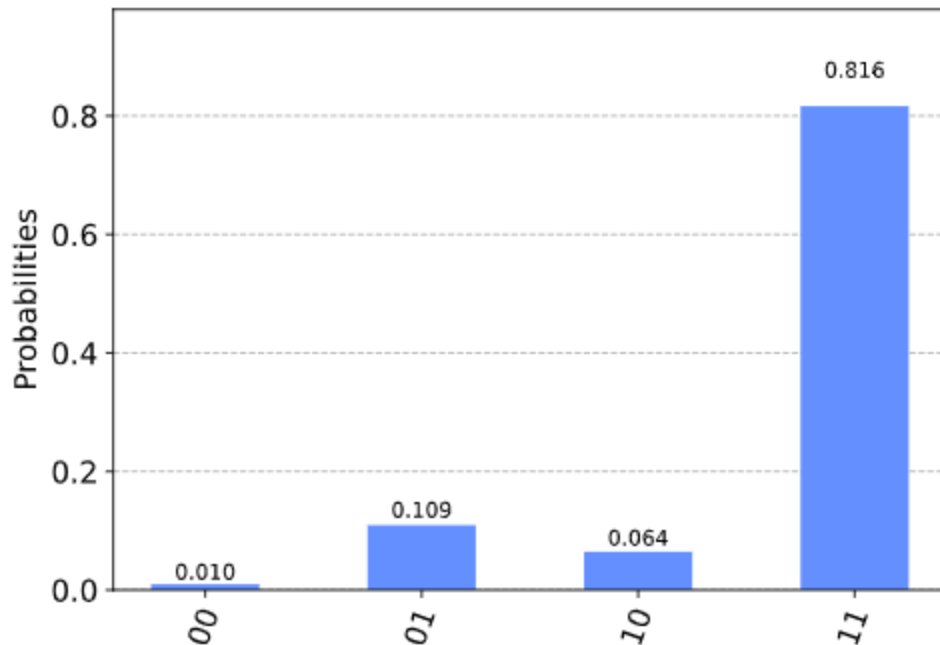


It can be clearly seen that this is a controlled Z-operation. Thus, for this particular case, our oracle is nothing but a controlled-Z gate.

The third step involves using the diffuser circuit such that at the output we get only $|11\rangle$ state with maximal probability. The final circuit is:



The results are shown as:

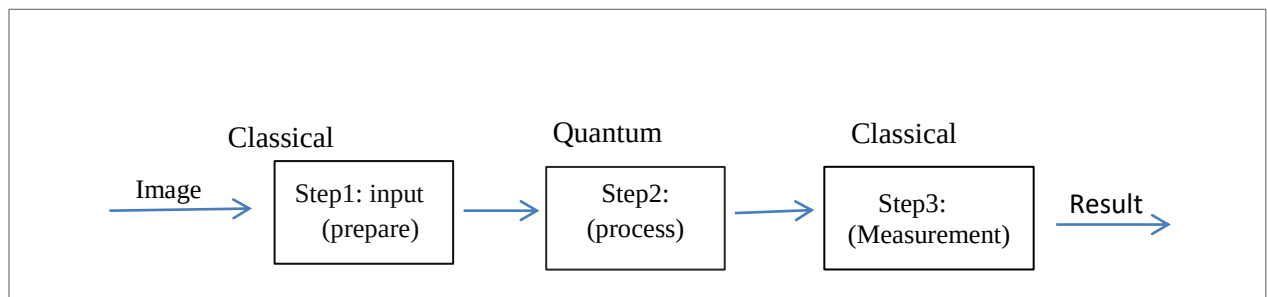


We confirm that when the measurement is carried out we get the state $|11\rangle$ is with maximum probability. The other results are due to errors in the quantum computation which can be substantially lowered.

Quantum Image processing

Quantum image processing is the converging of quantum computation and image processing. It's anything but a recently arising field and offers extraordinary freedoms as well as numerous incredible difficulties for analysts to foster more productive and practicable uses of quantum image processing[3]. It paly a fundamental job for the Air Force, just as different parts of the military and the business area. It is utilized in many key Air Force regions, like target following, and navigational purposes. Nonetheless, there are many wanted uses of image processing that isn't possible on a classical computer in light of the fact that, these sorts of assignments may require high computational force[7]. These may incorporate include tasks like modern model based objective obtaining utilizing an extremely enormous arrangement of information models[11]. Since quantum

computers hypothetically guarantee the capacity to perform computational undertakings a lot quicker than classical computers[12]. In this way, quantum computers might have the option to practically take care of a portion of these issues. In quantum picture preparing task the classical image data is first encoded into an appropriate quantum state and afterward processing tasks are finished offering the dramatic speedup during these processing tasks[13]. QIP can be possibly done in three stages as Fig.2 shows:



Fig(8) The framework of QIP.

Step(1): The first step starts with quantum image preparation which means storing image data into a quantum state. This step involves the classical-quantum interface. There are many existing methods of quantum image representation that have been developed over years. All these different representations have different preparation ways.

Step(2): The next step involves the processing task being carried onto the quantum image. This step is carried out by using a quantum computer which does the transformations on these quantum images. Various quantum algorithms have been proposed for these tasks which can be divided into several categories:

quantum geometric transformation, quantum color transformation, quantum image segmentation, quantum feature extraction, quantum image watermark and encryption, and so on.

Step(3): The third step is retrieval of the image data so that classical data is obtained by doing the measurement on this quantum state. This step is the quantum-classical interface.

11. Quantum image representation

Quantum image representation is the technique of how the classical images are encoded into a quantum image. There are various methods of quantum image representations that have been proposed by researchers [14]. Some of them are:

Qubit Lattice, Entangled image[5], Real Ket[6], Flexible Representation of Quantum images (FRQI)[21], Novel enhanced quantum image representation (NEQR)[22]. Among all these representations there are two important methods one is FRQI and the other is the advanced version of it known as NEQR. In FRQI method the information of an image is stored in single qubit state. FRQI can resolve the ongoing calculation issue of image processing applications. Be that as it may, on the grounds that FRQI utilizes just a single qubit to store the gray-scale data for every pixel in an image, some complex advanced image preparing tasks like certain perplexing color operations are impossible dependent on FRQI.

In NEQR model the information of an image is stored in the basis state of a qubit sequences. It uses two set of qubit sequences one to store the gray-scale value of the image and the other to store information related to the position of every pixel. Hence, to store the digital picture utilizing quantum mechanics, two entangled qubit sequences are utilized in NEQR to store the entire picture, which addresses the gray-scale and positional data of the relative multitude of pixels, as every basis qubit state is linearly independent free the image processing task turns out to be a lot simpler than FRQI and numerous operations should likewise be possible. Additionally, the time intricacy for setting up the NEQR quantum image displays an around quadratic decrease when contrasted with FRQI.

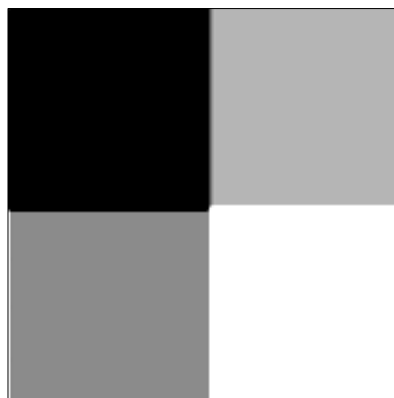
12.Qiskit

Quantum information science kit or Qiskit is an open source platform provided online by IBM for manipulating, writing quantum programmes and circuits[24]. The language of the Qiskit is based on Python's Jupyter notebook. The code of the program can be written in jupyter notebook and then send to a real Quantum computer or simulator provided online by IBM.

Image processing in Qiskit

Classical image

A image is made of pixels which are actually the small regions containing the information about the image. The primary colour scheme of an image is generally RGB whose different proportions in a pixel make all the colourful image that we see. Thus, an image has three dimensions length ,width and the base colour which is three for a colored image. The image used is a simple version of an image and is a (2x2) image having only four pixels and one color (i.e. grayscale) which is shown as in fig(9). Our task in is to locate the darkest (0 intensity black) pixel using grover search algorithm.



Fig(9) A classical 2x2 image

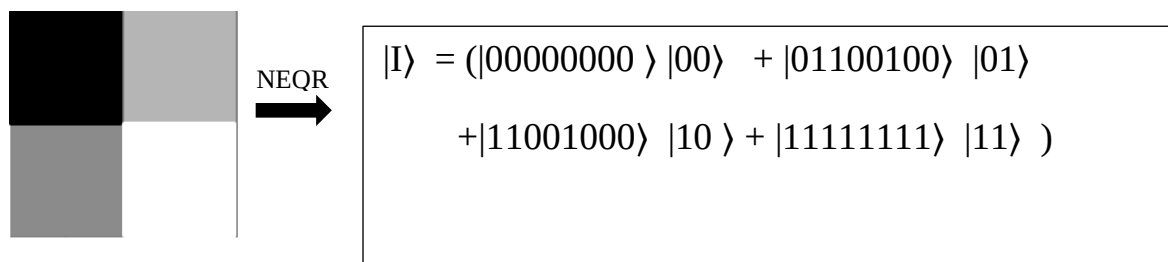
Step1: Image preparation.

NEQR offers much more advantages than NEQR so we will be using NEQR representation of the image . Our image is made up of four pixels so we will need 2 qubits for encoding position information[27]. For the color we will need 8 qubits because the grayscale intensity of an image can vary from 0 to 255. The bitstrings that encode the color for our image are shown in table(1)

Pixel position	Binary string	Grayscale intensity
$ 00\rangle$	$ 00000000\rangle$	0–Black
$ 01\rangle$	$ 01100100\rangle$	100–Darkshade
$ 10\rangle$	$ 11001000\rangle$	200–Lightshade
$ 11\rangle$	$ 11111111\rangle$	255–White

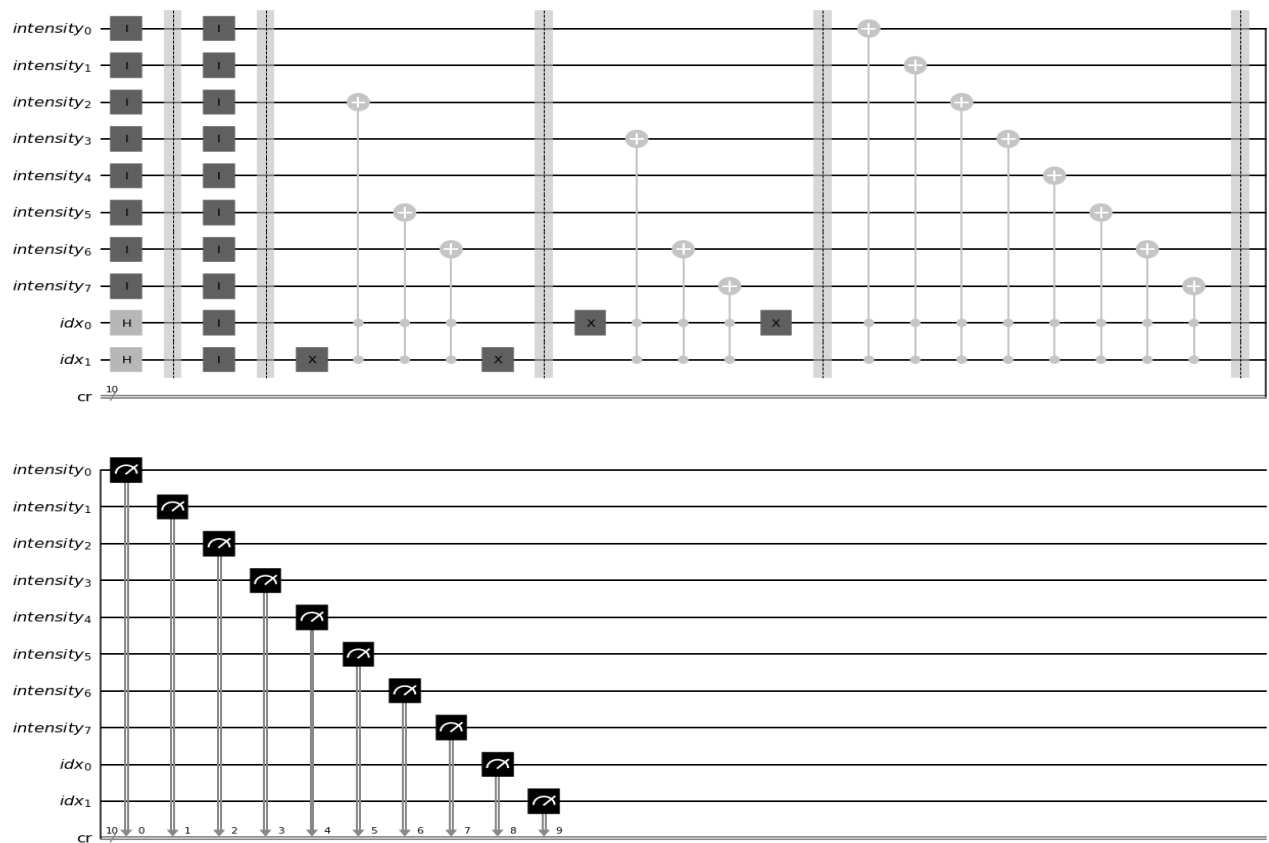
Table(1) Showing the binary equivalent of the color intensity values of the image.

So, in total we will need 10 qubits when initializing the system. The NEQR representation of the image can be done by the code (see Appendix). The image information is now encoded in an entangled state of two qubit sequences with one sequence containing information about grayscale intensity and the other with pixel position. The final state can be described as:



A (2x2) classical image and its representation NEQR.

The quantum circuit that does this encoding is shown in fig(10) as:



Fig(10) Quantum circuit of NEQR

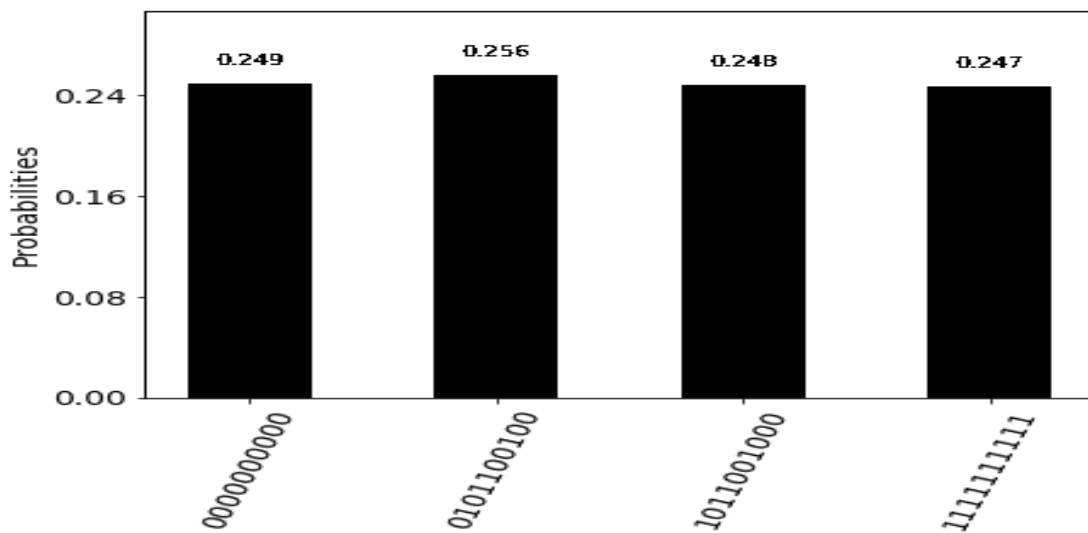
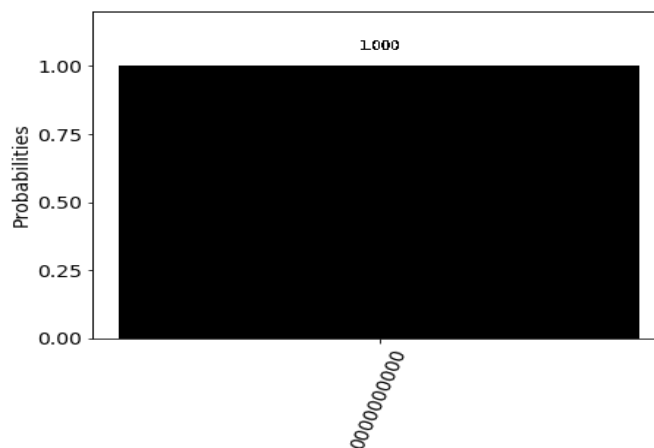


Fig10(b) Histogram showing results of image after encoding.

Step2: The pixel with minimum intensity value for our case is the first pixel located at (0,0). Since our final state contains a 10 qubit sequence we will use 10 qubit grover algorithm (for code See appendix) and tell it to search for the minimum intensity pixel. The output of the circuit will be a 10 qubit string which in our case is “0000000000” with maximal probability.

Step3: The most important part is the measurement part where we use 10 classical registers to extract the output from the circuit. After doing the measurement the output result is plotted as a histogram as shown in fig



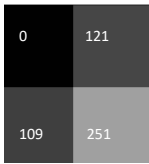
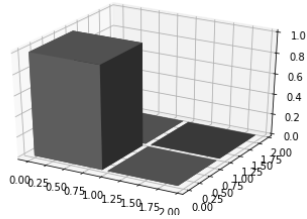
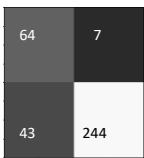
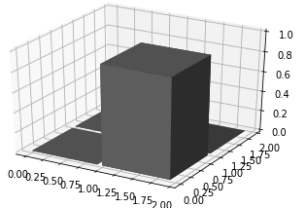
Fig(11) showing the Histogram results of the Grover algorithm.

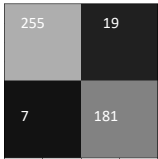
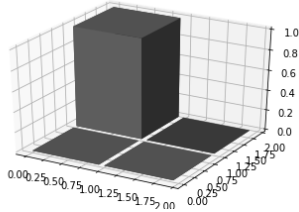
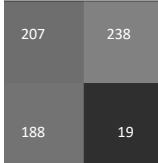
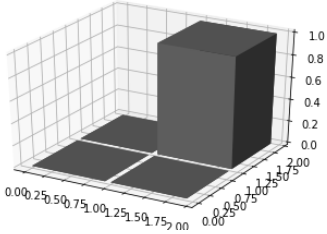
13. Cirq

Cirq is also an open source library provide by Google for creating and manipulating quantum circuits. Cirq uses different types of qubits like as line (linear array), grid qubits (two dimensional qubits) for storing and manipulating quantum information[31]. The language is written in Python and the circuits can be run on the quantum simulators provided at Google.

Image processing done in Google Cirq

In Cirq environment the information of the image is encoded by forming a (2x2) matrix. The code (see appendix) is written in Python's Jupyter notebook and run on Quantum simulators provided by Google. The different variations of a 2x2 image are as shown in table.2 is drawn at random with gray-scale intensity value ranging from 0-255. The image is then converted into a numpy array so pixels can be addressed uniquely based on their intensity values. A 2x2 matrix is obtained containing the intensity information of the pixels at their respective positions. Now our task is to find the pixel with the minimum intensity for this we will define x' (x_{prime} = index of the minimum intensity pixel of the image) such that our Grover search algorithm (for code see appendix) implements the oracle function $\{f(x) = 1 \text{ if } x == x' \text{ (winning state), } f(x) = 0 \text{ if } x \neq x'\}$. We use two types of Qubits for implementing Grover algorithm in Cirq. LineQubits from cirq which are identified by an integer index in a line and some ancilla qubits which are required to achieve some specific goals in computation (e.g reversible computation). After implementation the algorithm reveals index of minimum intensity pixel. The results are plotted on a 3d histogram as shown in Table.2

Sno	Image with intensiy values	Index of black pixel	3D Histogram
1		(0,0)	
2		(0,1)	

3		(1,0)	
4		(1,1)	

Table(2) variation of image and respective plots

Explanation of the results

In the above mentioned table(2) many different variations in intensities of the same image are shown. In first case the intensities of four pixels is as [0,121,43,244]. The pixel with the minimum intensity is at location (0,0) i.e it is the first pixel. Similarly the intensities of all other variations are shown. The results are plotted on a 3d histogram which gives a solid bar of height 1 for position of minimum intensity pixel and 0 for all other positions.

14.Summary and Conclusion

I started my journey in quantum computation from one of the documentaries of Brain Green on Quantum mechanics. I became fond of quantum computation and began to understand the quantum physics behind it. I started watching online lectures provided by Qiskit on the subject which helped me a lot in getting a firm grip on theoretical part as well as the coding part. For understanding mathematical basics regarding the subject i read book by David McMohan on quantum computation and for the theoretical part i got through the famous book on quantum

computation by Nelson and Chaung. This book helped me a lot in understanding how quantumcomputrs work. I then started to read about quantum algorithms. There are various algorithms but, the algorithm that got my attention was grover algorithm given by Lov Grover. This algorithm seemed to offer many advantages and i choose to use this algorithm for image processing. I then came to know about the different methods of quantum image representation like FRQI and NEQR from the online qiskit textbook which also provided the necessary coding details . I read both the approaches about quantum representation. Since, NEQR is enhanced version of FRQI i used NEQR in my processing task. I used online platforms like qiskit (provided by IBM) and Cirq (provided by google) for computational approach towards the task. I signed up for both the platforms so, i could get the real acess to quantum computers and play with quantum circuits . At the end of the journey i m now able to understand working of quantum computers and how these are implemented in doing real day to day problems. I am now in a poisitio to encode and communicate with quantum computers using different python languages. I hope to carry my future work ahead in the same field and search for next quantum breakthrough.

Using quantum mechanics and merging it with image processing is an effective way to address the high computational requirements of classical image processing. We have also shown how classical image is represented in both these frameworks, the first one uses NEQR in which the qubit sequences store the information and second one uses a matrix with elements representing the image information.

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16.Appendix

NEQR

```
# Importing standard Qiskit libraries and configuring account
import qiskit as qk
from qiskit import QuantumCircuit, Aer, IBMQ
from qiskit import transpile, assemble
from qiskit.tools.jupyter import *
from qiskit.visualization import plot_histogram
from math import pi
from qiskit.quantum_info import Operator
import numpy as np
import matplotlib as mp
from qiskit.aqua.algorithms import QSVM

import numpy as np
from qiskit import QuantumRegister, ClassicalRegister

# Initialize the quantum circuit for the image
# Pixel position
idx = QuantumRegister(2, 'idx')
# grayscale pixel intensity value
intensity = QuantumRegister(8, 'intensity')
# classical register
cr = ClassicalRegister(10, 'cr')

# create the quantum circuit for the image
qc_image = QuantumCircuit(intensity, idx, cr)

# set the total number of qubits
num_qubits = qc_image.num_qubits

#qc_image.draw()

# Initialize the quantum circuit

# Optional: Add Identity gates to the intensity values
for idx in range(intensity.size):
    qc_image.i(idx)
```

```

# Add Hadamard gates to the pixel positions
qc_image.h(8)
qc_image.h(9)

# Separate with barrier so it is easy to read later.
#qc_image.barrier()
#qc_image.draw()

# Encode the first pixel, since its value is 0, we will apply
ID gates here:
value00='01100100'
for idx in range(num_qubits):
    qc_image.i(idx)

#qc_image.barrier()
#qc_image.draw()

# Encode the second pixel whose value is (01100100):
value01 = '01100100'

# Add the NOT gate to set the position at 01:
qc_image.x(qc_image.num_qubits-1)

# We'll reverse order the value so it is in the same order whe
n measured.
for idx, px_value in enumerate(value01[::-1]):
    if(px_value=='1'):
        qc_image.ccx(num_qubits-1, num_qubits-2, idx)

# Reset the NOT gate
qc_image.x(num_qubits-1)

#qc_image.barrier()
#qc_image.draw()

# Encode the third pixel whose value is (11001000):
value10 = '11001000'

# Add the CNOT gates, where 0 is on X pixel:
qc_image.x(num_qubits-2)
for idx, px_value in enumerate(value10[::-1]):
    if(px_value=='1'):
        qc_image.ccx(num_qubits-1, num_qubits-2, idx)
qc_image.x(num_qubits-2)

```

```

qc_image.barrier()
qc_image.draw()

# Encode the third pixel whose value is (10101010):
value11 = '11111111'

# Add the CCNOT gates:
for idx, px_value in enumerate(value11):
    if(px_value=='1'):
        qc_image.ccx(num_qubits-1,num_qubits-2, idx)

qc_image.barrier()
qc_image.measure(range(10), range(10))
qc_image.draw()

qasm_sim = Aer.get_backend('qasm_simulator')
t_qc_image = transpile(qc_image, qasm_sim)
qobj = assemble(t_qc_image, shots=8192)
job_neqr = qasm_sim.run(qobj)
result_neqr = job_neqr.result()
counts_neqr = result_neqr.get_counts()
print('Encoded: 00 = 0')
print('Encoded: 01 = 01100100')
print('Encoded: 10 = 11001000')
print('Encoded: 11 = 1')

print(counts_neqr)
plot_histogram(counts_neqr)

```

Grover algorithm used in NEQR

```

def diffuser(n):

    # create a quantum circuit on n qubits
    qc = QuantumCircuit(n, name='Diffuser')

    ### WRITE YOUR CODE BETWEEN THESE LINES - START

    # apply hadamard gates to all qubits
    qc.h(range(n))

```

```

# call the phase oracle applied to the zero state
qc.append(phase_oracle(n, [0]), range(n))
# apply hadamard gates to all qubits
qc.h(range(n))

### WRITE YOUR CODE BETWEEN THESE LINES - END

return qc
def phase_oracle(n, indices_to_mark, name = 'Oracle'):

    # create a quantum circuit on n qubits
    qc = QuantumCircuit(n, name=name)

    ### WRITE YOUR CODE BETWEEN THESE LINES - START

    # create the identity matrix on n qubits
    oracle_matrix = np.identity(2**n)
    # add the -1 phase to marked elements
    for index_to_mark in indices_to_mark:
        oracle_matrix[index_to_mark, index_to_mark] = -1

    ### WRITE YOUR CODE BETWEEN THESE LINES - END

    # convert your matrix (called oracle_matrix) into an operator, and add it to the quantum circuit
    qc.unitary(Operator(oracle_matrix), range(n))

    return qc
def Grover(n, indices_of_marked_elements):

    # Create a quantum circuit on n qubits
    qc = QuantumCircuit(n, n)

    # Determine r
    r = int(np.floor(np.pi/4*np.sqrt(2**n/len(indices_of_marked_elements))))
    print(f'{n} qubits, basis states {indices_of_marked_elements} marked, {r} rounds')

    # step 1: apply Hadamard gates on all qubits
    qc.h(range(n))

```

```

        # step 2: apply r rounds of the phase oracle and the diffu
ser
        for _ in range(r):
            qc.append(phase_oracle(n, indices_of_marked_elements),
range(n))
            qc.append(diffuser(n), range(n))

        # step 3: measure all qubits
        qc.measure(range(n), range(n))

    return qc

mycircuit = Grover(n, [marked_bitstring])
mycircuit.draw()

#from qiskit import Aer, execute
#simulator = Aer.get_backend('qasm_simulator')
#counts = execute(mycircuit, backend=simulator, shots=1000).re
sult().get_counts(mycircuit)
#from qiskit.visualization import plot_histogram
plot_histogram(counts)

```

Cirq

```

import cirq
import sympy
import numpy as np
import seaborn as sns
import collections
import matplotlib.pyplot as plt
from cirq.contrib.svg import SVGCircuit
from PIL import Image
import random
import matplotlib.pyplot as plt
import numpy as np
from cirq import plot_state_histogram as plt_hist
image1=np.random.randint(255,size=(2,2))
plt.imshow(image1,cmap='gray',vmin=0,vmax=255)
plt.show()

```

```

plt.imshow('grayimg.jpeg', image1, cmap='gray')

FILENAME='grayimg.jpeg'
image2 = Image.open(FILENAME)
pixels=np.asarray(image2)
print(pixels)
"""Get qubits to use in the circuit for Grover's algorithm."""
# Number of qubits n.
nqubits = 3
# Get qubit registers.
qubits = cirq.LineQubit.range(nqubits)
ancilla = cirq.NamedQubit("Ancilla")
xprime = np.unravel_index(np.min(pixels), pixels.shape)
def make_oracle(qubits, ancilla, xprime):
    """Implements the function {f(x) = 1 if x == x', f(x) = 0
    if x != x'}."""
    # For x' = (1, 1), the oracle is just a Toffoli gate.
    # For a general x', we negate the zero bits and implement
    a Toffoli.
    # Negate zero bits, if necessary.
    yield (cirq.X(q) for (q, bit) in zip(qubits, xprime) if not
    bit)

    # Do the Toffoli.
    yield (cirq.TOFFOLI(qubits[0], qubits[1], ancilla))

    # Negate zero bits, if necessary.
    yield (cirq.X(q) for (q, bit) in zip(qubits, xprime) if not
    bit)
def grover_iteration(qubits, ancilla, oracle):
    """Performs one round of the Grover iteration."""
    circuit = cirq.Circuit()

    # Create an equal superposition over input qubits.

```

```

circuit.append(cirq.H.on_each(*qubits))

# Put the output qubit in the  $|-\rangle$  state.
circuit.append([cirq.X(ancilla), cirq.H(ancilla)])

# Query the oracle.
circuit.append(oracle)

# Construct Grover operator.
circuit.append(cirq.H.on_each(*qubits))
circuit.append(cirq.X.on_each(*qubits))
circuit.append(cirq.H.on(qubits[1]))
circuit.append(cirq.CNOT(qubits[0], qubits[1]))
circuit.append(cirq.H.on(qubits[1]))
circuit.append(cirq.X.on_each(*qubits))
circuit.append(cirq.H.on_each(*qubits))

# Measure the input register.
circuit.append(cirq.measure(*qubits, key="result"))

return circuit
"""Create the circuit for Grover's algorithm."""
# Make oracle (black box)
oracle = make_oracle(qubits, ancilla, xprime)

# Embed the oracle into a quantum circuit implementing Grover's algorithm.
circuit = grover_iteration(qubits, ancilla, oracle)
print("Circuit for Grover's algorithm:")
print(circuit)
"""Simulate the circuit for Grover's algorithm and check the output."""
# Helper function.
def bitstring(bits):

```

```

        return "".join(str(int(b)) for b in bits)

# Sample from the circuit a couple times.
simulator = cirq.Simulator()
result = simulator.run(circuit, repetitions=1000)
# Look at the sampled bitstrings.
frequencies = result.histogram(key="result", fold_func=bitstring)
print('Sampled results:\n{}'.format(frequencies))
# Check if we actually found the secret value.
most_black_bitstring = frequencies.most_common(1)[0][0]
print("\nMost black pixel: {}".format(most_black_bitstring))

```

Plotting 3d Histogram of results

```

image=np.array(image1)

a = image.min()

image = np.where(image == a ,1, image)

image = np.where(image != 1, 0, image)

print(image)

data_array = np.array(image)

fig = plt.figure()

ax = fig.add_subplot(111, projection='3d')

x_data, y_data = np.meshgrid( np.arange(data_array.shape[0]),
                                np.arange(data_array.shape[1]) )

x_data = x_data.flatten()

y_data = y_data.flatten()

z_data = data_array.flatten()

ax.bar3d( x_data,
           y_data,

```

```
np.zeros(len(z_data)),0.93, 0.93, z_data )  
plt.show()
```

